

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

CxJHdgavH9tvQn5UApwUDJABXYhh3W3QE4wvovasW9Dc

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Compounds: Diphenhydramine, Ergotamine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Diphen-ergotamine

Formula: C₃₉H₄₇N₅O₄

Weight: 649.83

Drug-likeness: 2.8

Activity:

Antihistamine and vasoconstrictor

Target Analysis

Histamine H1 receptor

Binding Affinity: -9.5

Histamine receptor antagonism to reduce allergic symptoms

Serotonin receptor (5-HT_{1B/1D})

Binding Affinity: -8.8

Moderate vasoconstriction used in migraine treatment

Toxicity Profile

Risk Level: High

Affected Systems: Heart, Liver, Central Nervous System

Mitigation Suggestions:

- Monitor cardiovascular function
- Use in monitored doses
- Combination with protective agents

Pharmacokinetics

Absorption:

Rapid but incomplete

Distribution:

Widely distributed due to lipophilicity

Metabolism:

Hepatic via CYP450 enzymes

Excretion:

Primarily renal

Half-life: 9-12 hours

Detailed Analysis

Diphen-ergotamine is a synthetic small molecule derived by combining the chemical properties of Diphenhydramine and Ergotamine. The intention behind this compound is to leverage the antihistamine activity of Diphenhydramine with the vasoconstrictor properties of Ergotamine to produce a novel agent capable of providing symptomatic relief in both allergies and migraine episodes. Diphen-ergotamine's potential mechanism involves blockade of the H1-histamine receptor to mitigate allergic responses, while simultaneously acting as an agonist on the 5-HT1B/D serotonin receptors to exert vasoactive effects beneficial in treating migraine headaches. Despite a promising dual role, caution is warranted due to associated risks including cardiovascular strain, sedation, and potential hepatotoxicity. The drug-likeness profile is favorable for further development, though a high lipophilicity index and possible CNS complications highlight the need for careful dose titration and formulation. A structured drug development timeline is outlined, with basic steps delineating molecular characterization to advanced clinical trials. Further research emphasis is placed on advancing safety profiling and extended application testing.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

<https://ALCHEMUS.ai>